

Memory-Gravity Attention: Persistent-Importance Retrieval for Long-Horizon Memory Tasks

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Abstract

We present Memory-Gravity Attention (MGA), a model-agnostic retrieval layer that augments similarity-based memory retrieval with persistent-importance signals. Standard approaches—including similarity-only and Generative-Agents-style scoring (recency + importance + similarity)—perform well on recency-friendly queries but fail on tasks requiring long-horizon persistent memory: retrieving facts stated once long ago (needle), tracking unresolved tasks (open-loop), and filtering noise. MGA introduces a multiplicative gate: $\text{score} = \text{sim} \times (1 + \sigma(\boldsymbol{\theta}^\top \mathbf{f}))$, where $\mathbf{f}$ comprises log-estimated persistent signals (recency, frequency, unresolvedness, utility, weight, goal-relevance) and $\boldsymbol{\theta}$ is learned via logistic regression on interaction-derived feedback. On a synthetic benchmark of 10 worlds (4,500+ nodes, 415 test queries, sentence-transformer embeddings), the Generative-Agents baseline achieves Recall@5 = 0.209 overall but scores 0.000 on needle, open-loop, and noise-resistance query families. MGA achieves non-zero recall on all three families (needle: 0.143, open-loop: 0.222, noise: 0.075) while maintaining competitive overall performance (Recall@5 = 0.195) and the lowest noise retrieval rate (Noise@5 = 0.017). We report learned feature coefficients showing that goal-relevance ($\theta = 2.92$) is the dominant persistent signal. Code and data are available at <https://github.com/jkdkr2439/Memory-Gravity-Attention>.

1 Introduction

Memory retrieval for language model agents is typically framed as a similarity search problem: given a query, retrieve the most semantically similar stored items. This approach works well when the answer is recent and phrased similarly to the query. However, it fails systematically on tasks that require *persistent importance*—information that remains relevant over long horizons regardless of recency or surface similarity.

Consider the following scenarios:

- A fact stated once in session 2 of 25, never repeated, but still valid and needed (needle).
- A task opened early and never closed, requiring recall despite many intervening events (open-loop).
- A constraint revised mid-project, where the old version has high similarity to the query but is stale (stale trap).
- A dense log with 40%+ chatter, where the system must retrieve signal from noise (noise resistance).

Park et al. [Park et al., 2023] proposed scoring memory as $\alpha \cdot \text{recency} + \beta \cdot \text{importance} + \gamma \cdot \text{relevance}$ for generative agents. This addresses some failure modes but, as we show empirically, still scores zero on needle, open-loop, and noise-resistance tasks in our benchmark.

We propose **Memory-Gravity Attention (MGA)**, which uses a multiplicative gate structure:

$$\text{score}_i = \text{sim}_i \times (1 + \sigma(\boldsymbol{\theta}^\top \mathbf{f}_i)), \quad (1)$$

where sim_i is the query-node cosine similarity, $\mathbf{f}_i$ is a vector of log-estimated persistent signals, and σ is the sigmoid function. The gate ensures that: (a) nodes with low similarity cannot be retrieved by importance alone, and (b) among similarly relevant nodes, persistent importance disambiguates.

2 Related Work

Memory for LLM agents. Park et al. [Park et al., 2023] introduced a three-term scoring function for generative agents. MemGPT [Packer et al., 2023] manages memory via a virtual context hierarchy. Reflexion [Shinn et al., 2023] uses self-reflection for episodic learning. None of these explicitly benchmark persistent-importance retrieval on long-horizon tasks.

Retrieval-augmented generation (RAG). Standard RAG [Lewis et al., 2020] uses dense retrieval (similarity-based). Recent work adds reranking [Glass et al., 2022], filtering, and hybrid search. These focus on factual QA, not persistent-importance memory management.

Attention and novelty. Predictive coding [Friston, 2010] and surprise-driven attention [Itti & Baldi, 2009] process prediction errors rather than similarities. Differential Transformer [Ye et al., 2024] subtracts two attention maps to reduce noise. Our work is complementary: MGA operates as an external retrieval layer, not a modification of transformer internals.

3 Method

3.1 Problem Setting

A memory store contains N nodes, each with content text, timestamp, and session ID. Given a query at time t , retrieve the top- k nodes that answer the query. Gold labels exist only in an oracle (evaluator-only) and are never visible to the retrieval system.

3.2 Signal Estimators

All signals are estimated from the interaction log without access to gold labels:

- **Recency:** $r_i = \exp(-(t_q - t_i)/\tau_r)$
- **Frequency:** $f_i = \log(1 + m_i)/\log(1 + m_{\max})$, where m_i counts near-duplicate mentions (embedding similarity > 0.75)
- **Unresolvedness:** binary heuristic detecting task-like nodes without subsequent closure matches

- **Goal relevance:** $g_i = \cos(\mathbf{e}_i, \mathbf{e}_{\text{goal}})$, normalized per retrieval call
- **Utility/Weight:** Beta-smoothed feedback from training episodes (default 0.5 = uninformative)

All features are min-max normalized to $[0, 1]$ per retrieval call.

3.3 MGA Gate Retriever

The feature vector (excluding similarity) is:

$$\mathbf{f}_i = [\text{recency}, \text{frequency}, \text{unresolved}, \text{utility}, \text{weight}, \text{goal_rel}] \in [0, 1]^6.$$

The MGA gate score is:

$$\text{score}_i = \text{sim}_i \times (1 + \sigma(\boldsymbol{\theta}^\top \mathbf{f}_i)). \quad (2)$$

$\boldsymbol{\theta} \in \mathbb{R}^6$ is learned via logistic regression on training queries (gold = positive, sampled non-gold = negative).

3.4 Baselines

- **Recency:** rank by r_i only.
- **Similarity:** rank by sim_i only.
- **GA:** $\alpha \cdot r_i + \beta \cdot w_i + \gamma \cdot \text{sim}_i$ with $\alpha = \beta = \gamma = 1$ [Park et al., 2023].
- **MGA linear:** $\boldsymbol{\theta}^\top [\text{sim}, \mathbf{f}]$ (ablation without gate structure).

4 Experimental Setup

4.1 Synthetic Benchmark

We generate 10 synthetic worlds, each with:

- 25 sessions, 10–25 events per session (~ 450 nodes/world)
- 3 domains: research paper, software project, design project
- 7 preference dimensions with 2–4 mid-log revisions (stale traps)
- 4–8 tasks with staggered open/close (open-loop tracking)
- 45% chatter events (noise)
- Template variation across 5–8 templates per event type

Query families: constraint recall, stale trap, open-loop, needle, noise resistance.

Train/test split: 40%/60% of queries per world. $\boldsymbol{\theta}$ learned on train only.

4.2 Embeddings

Sentence-transformers (all-MiniLM-L6-v2) for all similarity computations. Run on NVIDIA T4 GPU via Modal.com.

4.3 Metrics

Recall@5, nDCG@5, Stale@5 (fraction of retrieved nodes that are superseded), Noise@5 (fraction that are chatter). All metrics on test queries.

5 Results

5.1 Overall Performance

Table 1: Retrieval performance across 415 test queries (10 worlds).

Retriever	Recall@5	nDCG@5	Stale@5	Noise@5
Recency	0.029	0.085	0.000	0.364
Similarity	0.138	0.496	0.340	0.046
GA	0.209	0.697	0.000	0.070
MGA gate	0.174	0.562	0.265	0.029
MGA linear	0.195	0.614	0.205	0.017

GA achieves the highest overall Recall@5 and nDCG@5. However, GA’s stale advantage (Stale@5 = 0.000) is largely due to recency naturally filtering old (stale) nodes rather than explicit stale detection.

5.2 Per-Family Breakdown

Table 2: Recall@5 by query family. Families where persistent memory matters most.

Family	Recency	Sim	GA	MGA gate	MGA linear
Constraint	0.036	0.151	0.257	0.191	0.220
Needle	0.000	0.143	0.000	0.143	0.000
Open-loop	0.000	0.097	0.000	0.111	0.222
Noise resist	0.000	0.050	0.000	0.075	0.025
Stale trap	0.036	0.135	0.275	0.187	0.230

The critical finding: **GA scores 0.000 on needle, open-loop, and noise-resistance families.** These are precisely the tasks where persistent importance beyond recency and similarity is required. MGA achieves non-zero recall on all three.

5.3 Learned Feature Importance

Goal relevance dominates ($\theta = 2.92$): knowing what the system is currently working on is the strongest persistent signal for retrieval. Recency and frequency provide additional signal.

Table 3: Learned θ (MGA gate), mean $\pm$ std across 10 worlds.

Feature	Coefficient
Goal relevance	2.92 ± 0.87
Recency	0.67 ± 0.50
Frequency	0.40 ± 0.35
Unresolvedness	0.21 ± 0.29
Utility	-0.01 ± 0.01
Weight	-0.01 ± 0.01

Utility and weight are uninformative in this benchmark (insufficient training episodes to estimate them).

6 Discussion

6.1 Why GA Fails on Persistent Tasks

GA combines recency + importance + similarity additively. On needle queries (old facts), recency suppresses old nodes. On open-loop queries (unresolved tasks), the importance proxy (weight) is uninformative without explicit task-tracking. On noise queries, GA has no mechanism to distinguish signal from chatter beyond similarity.

MGA’s gate structure addresses this: the persistent features (especially goal-relevance and unresolvedness) amplify nodes that similarity alone identifies as marginally relevant.

6.2 Limitations

1. **Synthetic data:** Results are on template-based synthetic logs. Real-world interaction logs have greater lexical diversity, ambiguity, and implicit references.
2. **Small training signal:** With only 4–5 training queries per world, θ estimation is noisy. More training data would likely improve MGA.
3. **Utility/weight features uninformative:** The benchmark does not generate enough use-episodes for these features to become meaningful. A longer-running agent would accumulate this signal.
4. **No lock detection:** The attention-entropy diagnostic (a key theoretical contribution) is not yet benchmarked.
5. **Overall recall:** GA still wins overall. MGA’s advantage is specifically on long-horizon tasks.

6.3 Broader Significance

MGA operationalizes a principle from existential systems theory: a system should attend not only to what is *similar* to its current query, but to what is *persistently important* for its ongoing operation. This is analogous to gap-based attention in biological systems, where survival-relevant mismatches receive priority processing regardless of recency or surface similarity.

7 Conclusion

We introduced MGA, a retrieval mechanism that gates similarity by learned persistent importance. On a controlled benchmark, we showed that:

1. Standard approaches (similarity, GA) score zero on tasks requiring long-horizon persistent memory.
2. MGA achieves non-zero recall on all persistent-memory task families.
3. Goal-relevance is the dominant persistent signal ($\theta = 2.92$).
4. MGA has the lowest noise retrieval rate (Noise@5 = 0.017 vs GA’s 0.070).

MGA does not replace similarity-based retrieval. It fills the gap where similarity and recency fail: persistent, long-horizon, noise-resistant memory.

References

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